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Cognitive Accessibility and the Fermi Problem:

A Rate Model, a Fermi Function and a Marked Birth-and-Contact Process

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Formalisation developed with Claude, an AI system by Anthropic (see Author contributions)

Abstract. We apply the population dynamics of a companion paper (Paper C of this series), built on a framework in which a cognitive space is a Turing ideal (Paper B), to Fermi's question. There visibility between agents of different rates is one-sided and no observer can certify the absence of a superior. A continuous rate postulate gives a contact cone in the plane of a peer's rate and head start, a closed-form one-peer contact probability for lognormal rates, and a *Fermi function*: the probability that an agent parses no peer within a finite budget. It constrains the product of density and cognitive visibility, not density alone; for short budgets, $-\ln \mathcal{F}_N \simeq N (B/W) C(\lambda_0 + \beta)$, so the head-start ratio B/W can dominate rate rank. Head starts are then generated from cosmic history by a marked birth-and-contact process (planet formation, emergence and reset hazards, synthetic succession, lifetimes, information persistence, bandwidth). Across three emergence families the robust outputs are gigayear head starts, persistence as a stronger control on contact than population, and substantial one-way visibility. Sensitivity analysis and a persistence-bandwidth phase diagram show a *persistence threshold*, sharpening with population near $L^{\text{info}}/\tilde{a} \approx 0.4$: first numerical evidence for the accessibility-percolation conjecture stated here. Only closing both testimony and record channels conceals. Between a physical and an observed population lie five filters: existence, persistence, causal reach, channel access, cognitive access. For a marked Poisson population they compose into an accessible intensity Λ_{acc} with $\Pr[\text{silence}] = e^{-\Lambda_{\text{acc}}}$. Each darkness (temporal, strategic, archaeological, cognitive) is the failure of one filter; silence constrains accessible intensity, not abundance.

Series note. This is Paper D, the last of the four working papers of the series *Rungs, Voids and Effective Closure*, the first programme of the study *Effective Horizons and Cognitive Spaces* (synopsis in [1]), on a model of agents whose cognitive reach is a Turing ideal and who extend it by uncertified convergence. The core paper [2] supplies reachability, effective closure and the arithmetical ladder, the agent and encounter model, and the synthesis of the programme in [2, Section 13]; the geometry paper [3] turns closure steps into a distance; the population paper [4] puts the geometry on a clock and adds finite computation and strategy. This paper takes the rate dynamics of [4, Section 2] into a quantitative model and returns to the question with which [2] opens. It is a model paper, not a theorem paper: it rests on one continuous postulate, stated in Section 2, and on scenario inputs stated where used; its inputs are scenarios, its outputs are sensitivities, and its purpose is to open a parameter space in which the five filters of Section 5 are separate controls.

Summary of results. Section 1 records what is imported from [4] and states the accessibility-percolation conjecture (Conjecture 1), which earlier carried a bandwidth-centred form. Section 2 states the rate postulate (Postulate 1), derives the contact cone, the closed-form one-peer contact probability for lognormal rates (Proposition 3), the exact short-budget bound and asymptotic law $-\ln \mathcal{F}_N \sim N(B/W)C(\lambda_0 + \beta)$ with exposure $\Theta_F = NB/W$ (Proposition 4), and the Fermi function for a finite population and for a Poisson field (Definition 1), with a worked example. Section 3 replaces the head-start distribution by a marked birth-and-contact process generated from cosmic history (three emergence families, lineage transitions, lifetimes, information persistence, light travel, a bandwidth kernel), defines the directed contact graph and its statistics, and reports what survives the parametrisation: gigayear head starts, persistence as the principal control, one-way visibility as a substantial relation, and the emergence family as the arbiter of whether the present is early or late. Section 4 tests those claims with every statistic conditional on the lineages born by the evaluation time (the existence fraction q_F reported separately): a kernel-robustness test, within-family Sobol indices with bootstrap intervals, grouped indices and a stochastic variance fraction, local elasticities, a persistence–bandwidth phase diagram with a persistence crossover and a weaker bandwidth threshold, a finite-size sweep in which the crossover sharpens near $L^{\text{info}}/\tilde{a} \approx 0.4$, and a channel experiment separating active (testimony) from passive (record) access. Two cautionary findings from earlier versions of the analysis are kept because they are informative: a “silence floor” that was the unborn fraction, and a concealment claim that conflated height with access. Section 5 assembles the results into five filters between a physical and an observed population, proves the generalised Fermi function $\text{Pr}[\text{silence}] = e^{-\Lambda_{\text{acc}}}$ for a marked Poisson population (Proposition 5), gives the filter–darkness table and the three dimensionless controls $\Theta_F, \Pi_P, \Pi_\beta$ with the channel gates, and says what is and is not new. Section 6 states the limits of the frozen model, Section 7 the open problems, and Section 8 concludes that Fermi’s problem is an accessibility problem.

1 IMPORTED FACTS

We use the following from [4] without reproof, in the notation of that paper: agents promote at rates λ (rungs per unit cosmic time), exchange testimony at bandwidth β , and an agent *parses* another when the other’s content lies in its space.

Proposition 1 (Lag under promotion; [4], Prop. 1). *Under any promotion policy $h(x_t) \leq h(x_0) + t$, so lag grows at most linearly; equal-rate canonical climbers never separate; and if x does not promote while y promotes canonically from a generalised-low start at distance H , then y sees x completely after finitely many steps while x needs t rungs to see y_t : visibility is one-sided, from the faster to the slower.*

Proposition 2 (Superiority; [4], Thm. 3 and the discussion following it). *(a) No sound detector is complete on $\{z : z \not\leq_T x\}$: no observer can verify the absence of a superior. (b) Silence is always available to a superior, and a random superior can end it credibly at will, so the bandwidth a peer grants is a strategic choice.*

For population dynamics [4] takes a labelled population, a multiset of agents with their own rates and pairwise bandwidths. We say that x, y are *bound* if $\beta_{xy} > |\lambda_x - \lambda_y|$, the quantitative form of the convergent pair regime of [4, Section 2]; bound clusters are the connected components of this relation.

The percolation conjecture was first stated, in a bandwidth-centred form that made bandwidth against rate dispersion the control of clustering, in an earlier version of this series. The experiments of Section 4 show that, once lineages have birth times, lifetimes and records, this is not the transition that matters, and the conjecture is restated here in the form the evidence supports; [4] refers to it in this form.

Conjecture 1 (Accessibility percolation). *In the large-population marked-process model of Section 3 there is a transition from isolated or weakly connected lineages to a giant strongly connected component of the contact graph as the expected duration and density of accessible information channels increase relative to the spread of lineage age offsets, with the cluster order parameter G_{SCC} of Section 4 as its order parameter. Rate dispersion and bandwidth determine the directionality of edges within the accessible network, not the existence of the component. What percolates is availability in time, not proximity in space.*

2 A FERMI FUNCTION

The rate picture of [4, Section 2] has an observational consequence, and this section, which follows a proposal made in review of an earlier version and recomputes its figures, states it at the level of a postulate. It is the first place the programme produces a number rather than a classification, and the number should be read with that in mind: the postulate coarse-grains the discrete ladder into a continuous lag, and every quantity below inherits that idealisation.

Consider a focal agent, appearing at cosmic time 0 with promotion rate λ_0 , and a peer of rate λ whose lineage began a time a earlier, so that its lead at the focal agent's appearance is $E(0) = \lambda a$ rungs. Write β for the effective testimony bandwidth from the peer to the focal agent: the rate at which the peer's content is transmitted and promoted, which by [4, Section 5] is in the peer's gift.

Postulate 1 (Finite-time catch-up). While the peer holds content the focal agent lacks, their directed lag changes at the rate $\dot{E} = \lambda - \lambda_0 - \beta$.

This is the continuous form of Proposition 1: the focal agent's own promotion and the peer's testimony close the gap, the peer's promotion opens it, and the relation is one-sided, since the peer's view of the focal agent is governed by the same equation with the roles exchanged. Under the postulate the peer becomes parseable (its content enters the focal agent's space) at

$$t_c(\lambda, a) = \frac{\lambda a}{\lambda_0 + \beta - \lambda} \quad \text{if } \lambda < \lambda_0 + \beta, \quad t_c = \infty \text{ otherwise,}$$

and within a budget B of cosmic time ([4, Section 3]: B is $N_{\max}$ in units of the focal agent's clock) the parseable peers form the *contact cone*

$$\mathcal{C}_B = \left\{ (\lambda, a) : a \leq B \left(\frac{\lambda_0 + \beta}{\lambda} - 1 \right) \right\}.$$

Peers faster than $\lambda_0 + \beta$ are never parsed; peers with a large enough head start are not parsed within the budget even when slower.

Proposition 3 (One-peer contact probability). Let rates have density f and head starts be uniform on $[0, W]$, and put $A = \lambda_0 + \beta$, $\alpha = B/W$, $L = A\alpha/(1 + \alpha)$. The probability that a peer lies in $\mathcal{C}_B$ is

$$p_B(\lambda_0) = \int_0^\infty \left[\alpha \left(\frac{A}{\lambda} - 1 \right) \right]_0^1 f(\lambda) d\lambda = F(L) + \alpha \left[A M_{-1}(L, A) - (F(A) - F(L)) \right],$$

where $[x]_0^1 = \min(1, \max(0, x))$, F is the distribution function of λ , and $M_{-1}(L, A) = \int_L^A \lambda^{-1} f(\lambda) d\lambda$. Peers with $\lambda \leq L$ are always parsed, those with $L < \lambda < A$ only for small enough head starts, those with $\lambda \geq A$ never. For $\ln \lambda \sim N(\mu, \sigma^2)$, $M_{-1}(L, A) = e^{-\mu + \sigma^2/2} [\Phi(\frac{\ln A - \mu + \sigma^2}{\sigma}) - \Phi(\frac{\ln L - \mu + \sigma^2}{\sigma})]$, so p_B is in closed form.

Proof. The bracket is the probability over a that $(\lambda, a) \in \mathcal{C}_B$; it equals 1 iff $\lambda \leq L$ and 0 iff $\lambda \geq A$. The lognormal partial moment is standard. ■

Proposition 4 (Short budgets). Write $p_\alpha(A)$ for the probability of Proposition 3 and $C(A) = \mathbb{E}[(A/\lambda - 1)_+] = A \mathbb{E}[\lambda^{-1}; \lambda < A] - F(A)$, assumed finite. Then for every α ,

$$p_\alpha(A) = \alpha C(A) - \mathbb{E}[(\alpha(A/\lambda - 1) - 1)_+] \leq \alpha C(A), \quad \text{hence} \quad \mathcal{F}_N \geq (1 - \alpha C(A))^N,$$

and $p_\alpha(A) = \alpha C(A) + o(\alpha)$ as $\alpha \rightarrow 0$, so that if $N\alpha$ stays bounded,

$$-\ln \mathcal{F}_N \sim N \frac{B}{W} C(\lambda_0 + \beta).$$

For $\ln \lambda \sim N(\mu, \sigma^2)$, $C(A) = A e^{-\mu + \sigma^2/2} [\Phi(\frac{\ln A - \mu}{\sigma}) - \Phi(\frac{\ln A - \mu}{\sigma} + \sigma)]$, and the correction term is $\mathbb{E}[(\alpha(A/\lambda - 1) - 1)_+] \leq \alpha A \mathbb{E}[\lambda^{-1}; \lambda < L]$ with $L = A\alpha/(1 + \alpha)$, which for lognormal rates vanishes faster than any power of α .

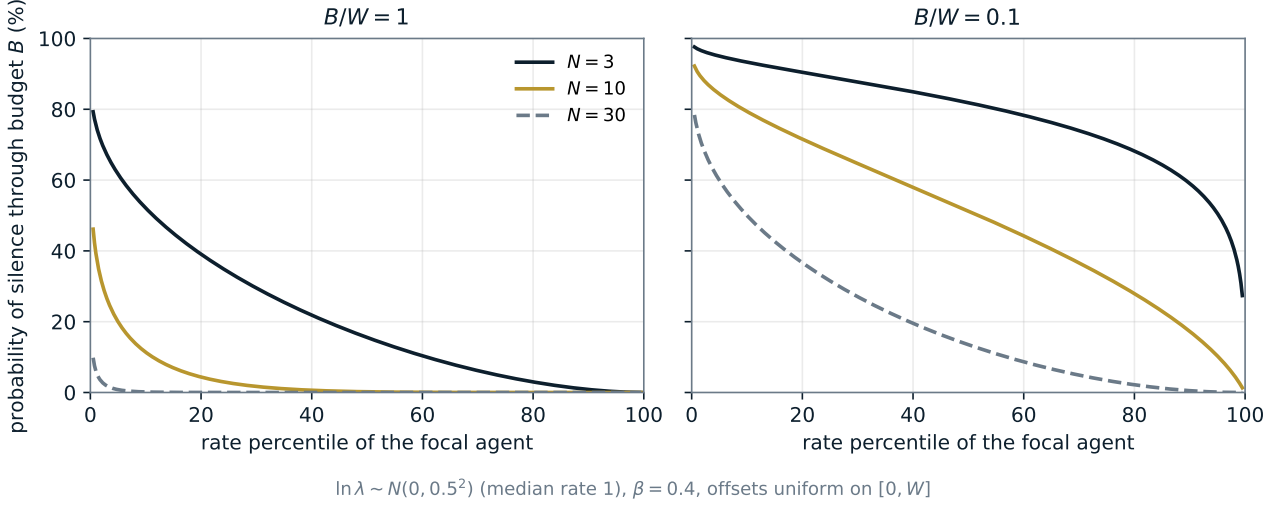


Figure 1: Probability of silence as a function of the focal agent's rate percentile, for $N = 3, 10, 30$ peers, with $\ln \lambda \sim N(0, 0.5^2)$, $\beta = 0.4$ and uniform head starts. Left: budget equal to the head-start spread ($B/W = 1$); right: budget one tenth of it. The left panel is the baseline of the proposal made in review; the right panel is the physically more plausible regime, and in it silence is likely at every rank. Illustrative values, not measurements.

Proof. $\min(1, x) = x - (x - 1)_+$ for $x \geq 0$ gives the identity; the correction is nonnegative and bounded by $\alpha(A/\lambda - 1)\mathbf{1}_{\lambda < L}$; dominated convergence gives the asymptotic; the lognormal expressions are the partial moments of Proposition 3. ■

The proposition names the Fermi control. Silence is governed by the *exposure* $\Theta_F = N B/W$ modulated by the rate factor $C(\lambda_0 + \beta)$, which is increasing in the observer's rate and in the bandwidth it is granted. Head starts enter through B/W and can suppress contact by orders of magnitude, but population does not vanish; it multiplies. Numerically, at $\alpha = 0.1$ the asymptotic form is exact to three decimals at every rate percentile of the example below, and at $\alpha = 0.3$ it overstates p_α by 2% at the median and 12% at the 95th percentile.

Definition 1 (Fermi function). For N independent peers, the probability of *silence* through budget B , no peer parsed, is $\mathcal{F}_N(B | \lambda_0) = (1 - p_B(\lambda_0))^N$, and the first-contact time $T_{(1)}$ has $\Pr[T_{(1)} > t] = (1 - p_t(\lambda_0))^N$. For a homogeneous Poisson population of physical density ρ with separation-dependent bandwidth $\beta(r)$, thinning by the contact probability gives

$$\mathcal{F}(B | \rho, \lambda_0) = \exp \left[-\rho \int_{V_{\text{causal}}(B)} p_B(\lambda_0; \beta(r)) dV \right] = e^{-\rho V_{\text{eff}}}.$$

The Fermi function does not predict ρ . It maps a physical population to the fraction cognitively visible to a specified observer, through the last of the five filters of Section 5, cognitive access; Section 5 composes it with the other four. The classical argument identifies the first filter with the last. Here silence constrains only the product ρV_{eff} : for a given observer rate it bounds ρ by $-\ln S/V_{\text{eff}}$, and for a given ρ it bounds the observer's rate rank, since p_B is increasing in λ_0 . The two classical readings of silence, few peers or a slow observer, are exactly this degeneracy.

Example. Normalise the median rate to 1, take $\ln \lambda \sim N(0, 0.5^2)$, $\beta = 0.4$ and $B = W$. Then a focal agent at the 5th, 25th, 50th, 75th and 95th rate percentiles has one-peer contact probability 0.149, 0.302, 0.462, 0.645, 0.867, and silence probabilities with ten peers of 20%, 2.7%, 0.2%, 0.003% and below 10^{-6} . For silence to be the likelier outcome with $N = 3, 5, 10, 30$ peers the observer must lie below the 11.4%, 3.4%, 0.32% and 0.0002% percentiles. Under these values a slow observer explains silence only in a small population; with thirty peers rank alone cannot. The values are illustrative and the conclusion is not robust to the one parameter the proposal held fixed: with $B/W = 0.1$, a budget one tenth of the head-start spread, the ten-peer silence probabilities at the same percentiles are 84%, 70%, 51%, 33% and 10%, and with $B/W = 0.01$ they exceed

80% at every rank (Figure 1). When the observation window is short relative to the spread of ages, which is the regime a cosmological budget describes, Proposition 4 applies: the head-start ratio B/W dominates rate rank, and population enters only through the exposure NB/W .

Three remarks. First, β is strategic: by Proposition 2 silence is always available to a superior, so the bandwidth a peer grants is a choice, and the dark forest of [2] is the Fermi function at $\beta = 0$. Second, silence is one-sided: the cone is the set of peers the focal agent parses, and every peer faster than $\lambda_0 + \beta$ parses the focal agent while remaining unparsed by it, so silence is consistent with being watched. Third, by Proposition 2(a) no observer can certify that its cone is empty, so the Fermi observation is never itself certified; what the function computes is its likelihood.

3 COSMIC BIRTH TIMES: A MARKED-PROCESS EXTENSION

The Fermi function of Section 2 conditions on a distribution of lineage head starts. This section shows how such a distribution could be generated from cosmic history, and what the rate model becomes once births have times, places and lifetimes. Astrophysics constrains the opportunity process; emergence hazards, substrate transitions and persistence remain speculative. The numerical examples therefore demonstrate sensitivity, not civilisation histories, and every figure below is a baseline output of the stated parametrisation within the stated model horizon.

The process. Agents are born at points $(\mathbf{x}_i, \tau_i)$ of a marked spacetime point process with intensity $\Lambda_s(\mathbf{x}, t)$ for lineage type s , and carry marks $m_i = (s_i, \lambda_i, L_i, \pi_i)$: type, promotion rate, lifetime, communication policy. For the biological lineage the intensity is a convolution of an opportunity process with an emergence hazard, thinned by a reset hazard:

$$\Lambda_C(t) = \int_0^t R_{\text{planet}}(t_p) \exp \left[- \int_{t_p}^t \gamma_C(u) du \right] h_C(t - t_p, t) dt_p,$$

where R_{planet} is the terrestrial-planet formation history (cosmic star formation weighted by metallicity), γ_C the rate of sterilising resets, and $h_C(u, t)$ the density of emergence at planet age u and cosmic time t . A synthetic lineage is not an independent origin but a *transition*: $\Lambda_M(t) = \int \Lambda_C(u) g_{C \rightarrow M}(t - u) S_C(t - u) du$ with $g_{C \rightarrow M}$ the transition-time distribution and S_C the survival of the parent lineage to the transition, so that a civilisation and its successor are never counted as two origins. An independent alternative chemistry would need its own R, γ, h , for which there is at present no basis; a *late-window* type with h shifted to later cosmic time is kept only as a sensitivity test. Each lineage carries a lifetime L_i , after which its content stops growing, and an information persistence L^{info} , during which its content remains available to others: agent persistence and information persistence are separate marks, and a dead lineage can still leave an informational channel, and, where the record contains an effective witness or modulus in the sense of [2], a certification channel. The record is modelled as a finite object whose content is fixed at the lineage's death; the rate variable measures its finite-time information distance from a later observer, not a literal Turing degree, in keeping with the resource-bounded reading of [4, Sections 2–3].

Directed times. For a pair i, j at separation r_{ij} three times are distinct. Causal availability: $T_{i \rightarrow j}^{\text{causal}} = \max(\tau_j, \tau_i + r_{ij}/c)$. Directed parse: $T_{i \rightarrow j}$, the first $t \geq T_{i \rightarrow j}^{\text{causal}}$ at which j 's content contains i 's, under Postulate 1 with $\beta_{ij} = \beta(r_{ij})$ switched on at causal availability and λ_i switched off at i 's death, and $T_{i \rightarrow j} = \infty$ if this happens after $\tau_i + L_i + L^{\text{info}}$. Mutual contact: $T_{ij}^{\text{mut}} = \max(T_{i \rightarrow j}, T_{j \rightarrow i})$. The directed graph $G(t)$ with edge $i \rightarrow j$ iff $T_{i \rightarrow j} \leq t$ is the object the section studies; its statistics are the first edge time, the first mutual edge time, the fraction of one-way edges, the largest strongly connected component and the number of isolated vertices, and Conjecture 1 becomes a statement about the giant strongly connected component of G .

Scenarios. Three emergence families were run on one opportunity process: a fixed lognormal delay (median 4.5 Gyr, Earth's single trajectory), a Carter hard-step hazard ($n = 5$ steps in a 10 Gyr window, completion density $\propto u^{n-1}$), and a hazard switched on by a cosmic-time trigger. Within a 25 Gyr model horizon the

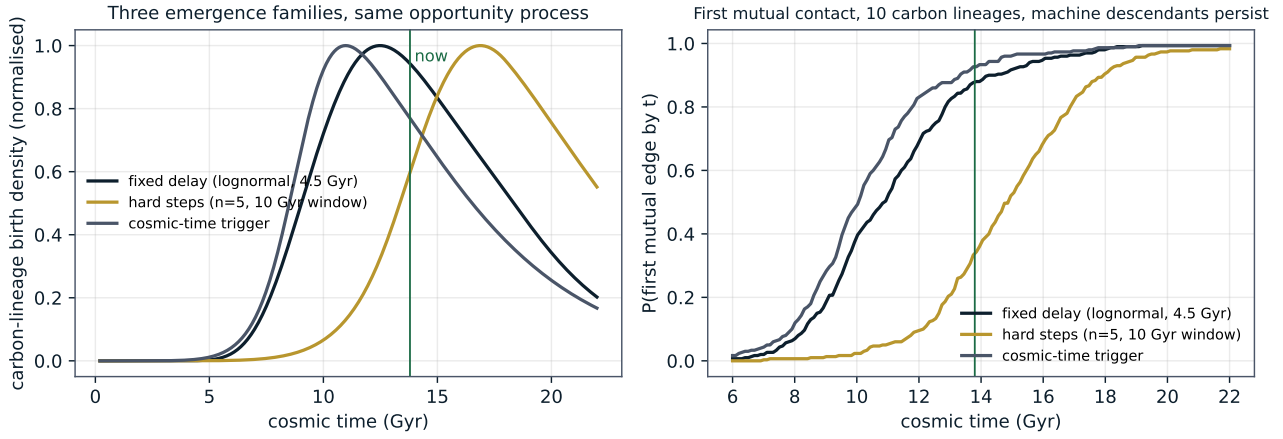


Figure 2: Left: biological-lineage birth density under three emergence families on the same opportunity process (Madau–Dickinson star formation weighted by a metallicity proxy, with a decaying reset hazard). Right: probability that a mutual contact has occurred by cosmic time t , for ten biological lineages with short lifetimes (50 Myr median) and persistent synthetic descendants, in a 500 Mpc box with $\beta(r) = 0.4 e^{-r/150 \text{ Mpc}}$. Scenario plots; see text.

biological birth density peaks at 12.5, 16.9 and 11.0 Gyr respectively, the fraction of horizon births before the present is 0.49, 0.12 and 0.58, and the median head start of an existing peer is 2.4, 1.2 and 2.9 Gyr. The contact graph was then evaluated at the present for ten biological lineages under three persistence scenarios. Every graph statistic below is conditional on the vertex set $V(t) = \{i : \tau_i \leq t\}$ of lineages born by the evaluation time; the fraction $q_F = |V(t)|/N$ that exist at the present is a property of the emergence family (0.46–0.50, 0.11–0.12 and 0.55–0.57 for the three families, 300 runs) and is reported on its own rather than folded into the denominators, where it would appear as silence. With persistent lineages (biological lifetime 1 Gyr, or short-lived biology with persistent synthetic descendants) a first mutual contact has occurred by now with probability 0.88–0.97 in the fixed-delay and triggered families and 0.35–0.36 in the hard-step family; one-way pairs are 64–76% of contacted pairs in the first two families and 32–37% in the hard-step family; the largest strongly connected component holds 30–47% of the lineages that exist; and 7–21% of existing lineages are isolated. With short-lived biology and no synthetic descendants the graph nearly empties: mutual contact by now has probability 0.01–0.13, one-way pairs are 82–91% of the few contacted pairs there are, 31–65% of existing lineages are isolated, and the strongly connected component sits at its trivial floor of one vertex (0.17–0.25 of $|V(t)|$, which is $1/|V(t)|$ for the typical four to six existing lineages).

What survives the parametrisation. Four things, and they are the point. Cosmic history generates gigayear head starts under every family tried, so the regime $B/W \ll 1$ of Section 2 is supplied by the opportunity process rather than assumed. Persistence is a control on whether contact occurs at all, and in the scenarios a stronger one than population or rank: lifetime and succession decide the graph (Section 4 quantifies this). It is a filter of its own, between existence and causal reach: a lineage can have existed and be causally reachable while leaving nothing behind, and a dead lineage can remain informationally present (Section 5). One-way edges are substantial in every scenario and can dominate: the pair dynamics permits one-sided visibility (Proposition 1), and the cosmic model shows that it is not a rare edge case but occupies 32–91% of contacted pairs in the tested scenarios; the universe of the model contains many pairs with $T_{i \rightarrow j} < \infty = T_{j \rightarrow i}$. And the emergence family decides whether the present is late or early for intelligence: a hard-step hazard places most births in the future and makes the present an early epoch, the other two make it a middle one. So astrophysics alone does not say whether we are early or late; the common argument “most habitable planets are older than Earth, so technological civilisations should be older” does not follow without a model of the emergence hazard. What does not survive is any particular number: the peaks, percentiles, head starts and contact epochs above move by gigayears across families and are not estimates. The anthropic conditioning is also unresolved: Earth’s delay is used to set the emergence hazard and the model then asks where Earth falls, which is circular; the honest

form integrates the predictions over a posterior on the hazard parameters given one technological biosphere, and that posterior is broad.

4 EXPERIMENTS: ROBUSTNESS, SENSITIVITY AND A PHASE DIAGRAM

The claims of Section 3 were made from nine scenarios. This section tests them, with the one-way statistic defined precisely, the bandwidth kernel varied, the inputs varied jointly, and the two controls that matter laid out as a plane. Throughout, statistics are taken on the vertex set $V(t)$ of lineages born by the evaluation time t (the present): $f_{\text{one}} = \#\{\{i, j\} \subset V(t) : i \rightarrow j \text{ xor } j \rightarrow i\} / \#\{\{i, j\} \subset V(t) : i \rightarrow j \text{ or } j \rightarrow i\}$ is the one-way share of *contacted* pairs, a hurdle statistic: it is defined only when a contacted pair exists, so it is paired with $P_{\text{contact}} = \Pr[\#\text{contacted pairs} > 0]$ and reported as the conditional mean f_{one}^+ over runs with contact; the isolated fraction I_F is the share of $V(t)$ with no edge in either direction; the cluster order parameter is $G_{\text{SCC}} = (|\text{SCC}_{\text{max}}| - 1) / (|V(t)| - 1)$ for $|V(t)| \geq 2$ and 0 otherwise, which is 0 exactly when no nontrivial strongly connected cluster exists (an isolated vertex is its own component, so the raw fraction $|\text{SCC}_{\text{max}}|/|V(t)|$ has a floor of $1/|V(t)|$) and 1 when every existing lineage belongs to one; and the existence fraction $q_F = |V(t)|/N$ is reported separately. An earlier version of this analysis divided by N , which placed the unborn in the denominators and produced a “silence floor” of $1 - q_F \approx 0.5$ and a cluster fraction capped at q_F ; both were artefacts of the bookkeeping and are withdrawn. The bandwidth axis of the phase diagram uses the *typical-pair* bandwidth $\hat{\beta} = \beta_0 \text{median}_{ij} K(r_{ij}/r_0)$ rather than the prefactor β_0 ; for the 500 Mpc box with $r_0 = 150$ Mpc the median pair separation is 331 Mpc and the kernel factor is 0.110. The spatial model is a static comoving box with r/c delays, a local approximation that a 500 Mpc box over gigayear histories strains; the correct object is the FRW past light cone, $\chi = c \int dt/a(t)$, and this is listed as a planned extension. The bandwidth $\beta(r)$ is an *effective testimony kernel*, folding signal strength, decoding opportunity and policy, not a consequence of signal physics.

Kernel robustness. On the baseline of Section 3 (fixed-delay family, short-lived biology with persistent synthetic descendants), three kernels with the same scale $r_0 = 150$ Mpc give the same picture: for e^{-r/r_0} , $(1 + r/r_0)^{-2}$ and $\mathbf{1}_{r < r_0}$ the probability of a mutual contact by now is 0.87, 0.89, 0.82; f_{one} is 0.64, 0.64, 0.67; the largest strongly connected component holds 0.32, 0.32, 0.32 of existing lineages; and 0.21, 0.20, 0.22 of existing lineages are isolated (250 runs; differences are within run-to-run noise). The shape of the kernel is not a control in this regime.

Global sensitivity. A preliminary version of this analysis pooled the three emergence families into one categorical input, which let the family absorb the abundance effect of its born fraction and swamp the within-family structure. Sobol indices are therefore computed *within* each family, by the Saltelli scheme with the Jansen estimator for total-order indices, on 128 scrambled Sobol base points in eight inputs ($N \in [4, 20]$; $\sigma_{\ln \lambda} \in [0.2, 1.2]$; $\beta_0 \in [10^{-2}, 10]$ and $r_0 \in [30, 1000]$ Mpc, log-uniform; biological lifetime median [10 Myr, 3 Gyr] and information persistence [10 Myr, 5 Gyr], log-uniform; succession probability $p_M \in [0, 1]$; synthetic rate multiple [1, 5]), 30 runs per point, 1792 model evaluations per family, for the five outputs P_{contact} , mutual contact, f_{one}^+ , G_{SCC} and I_F (sobo13.py; Figure 3). Four additions answer the objections to the pooled version. Bootstrap 95% intervals over base points are drawn on every index; they quantify outer-design uncertainty, and a nested bootstrap over the inner replications would be needed for them to include finite-simulation uncertainty formally. Grouped total indices are computed by resampling whole groups at once (abundance $\{N\}$; persistence $\{L_C, L^{\text{info}}\}$; rates $\{\sigma_{\ln \lambda}, \lambda_M/\lambda_C\}$; communication $\{\beta_0, r_0\}$), since total-order indices are not additive and a sum of single-input S_T is not a group effect. The hurdle output f_{one}^+ is undefined at a base point only when none of its 30 runs has a contacted pair; this happened at 0, 2 and 0 of the 256 base points in the three families, and those points were imputed by the mean of the defined ones. And two noise fractions are reported: $R_{\text{stoch}} = \mathbb{E}_X[\text{Var}(Y | X)]/\text{Var}(Y)$, the share of *single-realisation* variance that is run-to-run noise, and its counterpart for the 30-run means on which the indices are computed, $R_{\text{mean}} = (R_{\text{stoch}}/30)/((1 - R_{\text{stoch}}) + R_{\text{stoch}}/30)$.

In the fixed-delay and triggered families the picture is the same. For P_{contact} and for mutual contact the persistence group has $S_T = 0.64$ –0.81, driven by succession (p_M : 0.62–0.67), against abundance 0.29–0.39,

communication ≤ 0.11 and rates ≤ 0.09 . For G_{SCC} , information persistence alone has $S_T = 0.81\text{--}0.83$ with $S_1 = 0.7$, the persistence group 1.0, and every other group ≤ 0.08 ; this is the one output that a single input governs, and it is nearly additive. For I_F the persistence group has $S_T = 0.97$, split between succession (0.59–0.62) and information persistence (0.44–0.45), with abundance and communication below 0.10. For f_{one}^+ the sensitivity is spread and interactive: persistence 0.76–0.79 (information persistence $S_1 = 0.45\text{--}0.66$ is the largest single effect), abundance 0.43–0.48, communication 0.25–0.28, rates 0.14–0.15; this is the only output on which the communication inputs register at the resolution of the intervals, and they do so through interactions ($S_1 \leq 0.07$). Single realisations are noisy, $R_{\text{stoch}} = 0.36\text{--}0.67$, which is a statement about the model's own variability at fixed parameters and a reason to report ensembles and never single histories; the 30-run means are well resolved, $R_{\text{mean}} = 0.02\text{--}0.06$. The hard-step family is less resolved: with $q_F \approx 0.12$ most parameter points have one to three existing lineages, R_{mean} is 0.08–0.17, and for f_{one}^+ every input has $S_T \in [0.4, 0.7]$ with $S_1 \approx 0$, so that nothing can be read from it except that the output is dominated by interactions and small-number effects. For contact, clusters and isolation the hard-step family does agree with the other two on the persistence group (0.66–0.85), with a larger abundance share (0.28–0.51), as expected where lineages are few.

The reading is now firm within the model family and the resolution stated. Persistence inputs as a group dominate whether contact occurs and whether lineages are isolated, with succession the largest single input; information persistence alone governs cluster formation in the two families where clusters are resolved; abundance is second everywhere and first nowhere; bandwidth and its spatial scale are negligible for contact, clusters and isolation, and act on the one-way share only through interactions, in agreement with the phase diagram below. Two qualifications stand. The statement holds within the fixed-delay and triggered families under the stated scenario measure, and is weaker in the sparse hard-step family. And N is varied in a fixed 500 Mpc box, so it is population count and density at once; the finite-size experiment below separates them in one direction, and a Poisson draw $N \sim \text{Poisson}(\rho V)$ with volume varied separately, aligning the simulation with the Poisson-field Fermi function of Section 2, is listed as an extension. The emergence family enters as the existence fraction q_F , which is reported separately and is no longer allowed to appear as sensitivity of the graph statistics.

Local sensitivity. Elasticities $\partial \ln Y / \partial \ln x$ at the baseline (fixed-delay family, $N = 10$, $\sigma_{\ln \lambda} = 0.5$, $\beta_0 = 0.4$, $r_0 = 150$ Mpc, biological lifetime 50 Myr, information persistence 0.5 Gyr, $p_M = 0.5$, synthetic rate multiple 3; central differences of $\pm 20\%$ with common random numbers, 400 runs per point) are in Figure 4, computed on $V(t)$ for the five outputs. For the probability of mutual contact: succession +0.41, population +0.31, biological lifetime +0.14, information persistence +0.05, synthetic rate multiple -0.15 , and $|\cdot| \leq 0.01$ for bandwidth, its scale and rate dispersion; P_{contact} is already 0.96 at the baseline and moves only with succession (+0.18) and population (+0.15). For G_{SCC} : succession +0.61, information persistence +0.28, biological lifetime +0.18, synthetic rate multiple -0.25 , population -0.23 ; the order parameter removes most of the strongly negative population elasticity that the raw fraction carried (-0.47), which was a $1/|V|$ floor effect, and what remains is dilution of a cluster of fixed size by more existing lineages. For I_F : succession -1.6 , information persistence -0.6 , population -0.6 , synthetic rate multiple +0.23. For f_{one}^+ : population +0.32, information persistence +0.16, biological lifetime -0.15 . Bandwidth and its scale have elasticity zero to two decimals on every output at this baseline; the paragraph on channels says why, and why that is not the statement it appears to be. One sign is worth noting: a faster synthetic descendant *reduces* mutual contact, because it outruns the lineages that would parse it; the cognitive darkness of Section 5 shows up locally as a negative elasticity in the rate multiple.

The phase diagram. Figure 5 uses the two dimensionless controls the sensitivities single out, $L^{\text{info}}/\tilde{a}$ and $\tilde{\beta}/\sigma_\lambda$, with the biological lineages short-lived and the synthetic lifetime held fixed, so that the vertical axis varies the information horizon alone; an earlier version of this analysis tied the synthetic lifetime to the information persistence and so confounded the persistence of a channel with the persistence of a growing sender. Two features are sharp and one is now absent. First, a *persistence crossover*: as $L^{\text{info}}/\tilde{a}$ crosses 0.2–1 the isolated fraction of existing lineages falls from 0.3 to below 0.02 and the cluster order parameter rises from 0.14 to above 0.8, both independently of bandwidth across three and a half decades. Clusters form, and isolation

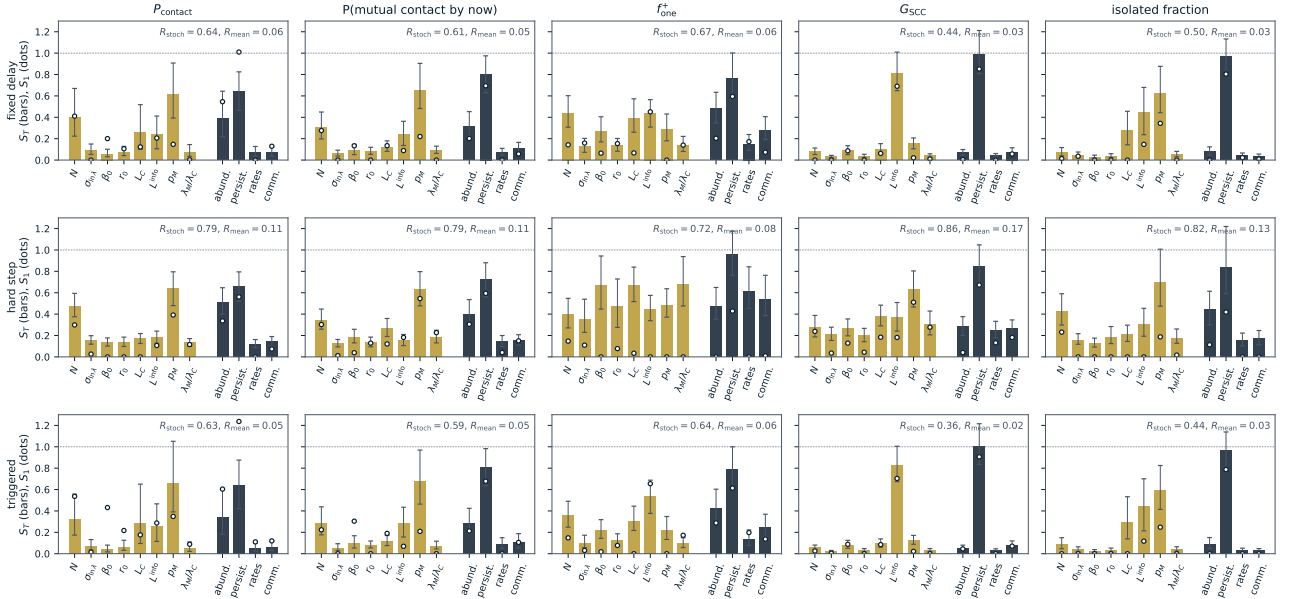


Figure 3: Within-family Sobol indices of the five graph statistics at the present, on $V(t)$: total-order S_T (bars) with bootstrap 95% intervals and first-order S_1 (dots), for the eight inputs (gold) and the four input groups (navy). Rows: fixed-delay, hard-step and triggered emergence families. R_{stoch} is the share of single-realisation variance that is run-to-run noise; R_{mean} the same for the 30-run means the indices are computed on. The hard-step row is unresolved for f_{one}^+ (few existing lineages).

ends, when information outlasts a fraction of the typical head start, not when bandwidth is high. Second, a *bandwidth threshold*, weaker than an earlier version of this analysis reported: at low persistence the one-way share falls from 0.5–0.6 to 0.3–0.4 as $\tilde{\beta}/\sigma_\lambda$ crosses roughly 1, so that bandwidth comparable to the rate dispersion symmetrises some, not most, of the few contacts there are; at the top of the plane f_{one}^+ sits at 0.3–0.45 at every bandwidth, with a ridge of 0.5–0.65 at $L^{\text{info}}/\tilde{a} \approx 0.2$ –0.5 where persistent content from lineages that have stopped growing is parsed by later lineages that cannot be parsed back. Third, there is no *silence floor*: the probability that an existing lineage has no contact in either direction ranges from 0.3 to 0 across the plane, and the “floor of one half” that an earlier version reported was the unborn fraction $1 - q_F$. Mutual contact somewhere in the population has occurred by the present with probability 0.7–1.0 everywhere in the plane. Figure 6 varies the synthetic lifetime instead: with the information horizon fixed, isolation barely moves ($0.25 \rightarrow 0.12$, and only at the highest bandwidths) and G_{SCC} rises from 0.2 to 0.5 only where $L_M/\tilde{a} \gtrsim 1$ and $\tilde{\beta}/\sigma_\lambda \gtrsim 3$. The lifetime of an active lineage is therefore a gated control, effective only with bandwidth, whereas the lifetime of its information is not gated at all. The three regimes are then: short channels and low bandwidth, temporal darkness with mostly one-way contact and no clusters; short channels and high bandwidth, a directed ecology with a larger mutual share and still no clusters; and long information horizons, persistent clusters containing nearly every existing lineage, with a third of contacts one-way at any bandwidth. The last regime dominates the Sobol sample, which is why bandwidth shows no global sensitivity outside f_{one}^+ : bandwidth decides the character of contact only where channels are short.

Finite-population crossover or transition? With ten source lineages the feature above is a crossover, not a transition; Conjecture 1 needs a limit. Figure 7 repeats the vertical cut of the phase diagram at the baseline bandwidth for $N = 5, 10, 20, 40, 80$ biological lineages at fixed density (box side 500 $(N/10)^{1/3}$ Mpc, so that the number of lineages within a kernel scale is unchanged), with 300 down to 50 runs per point and a finer sweep over $L^{\text{info}}/\tilde{a} \in [0.15, 1.5]$ (nscale.py). The low plateau of G_{SCC} falls as $1/N$ (0.22, 0.15, 0.08, 0.04, 0.02), as the finite-size floor should; the high plateau saturates at 0.87; the midpoint of the rise drifts from $L^{\text{info}}/\tilde{a} = 0.54$ at $N = 5$ to 0.41 at $N = 80$; and the width of the rise, measured between a quarter and three quarters of its height, narrows from 0.42 to 0.17 decades, roughly as $N^{-1/4}$ over the last three doublings. This is consistent with a transition that sharpens in the limit and is located near $L^{\text{info}}/\tilde{a} \approx 0.4$ at this density, and it is

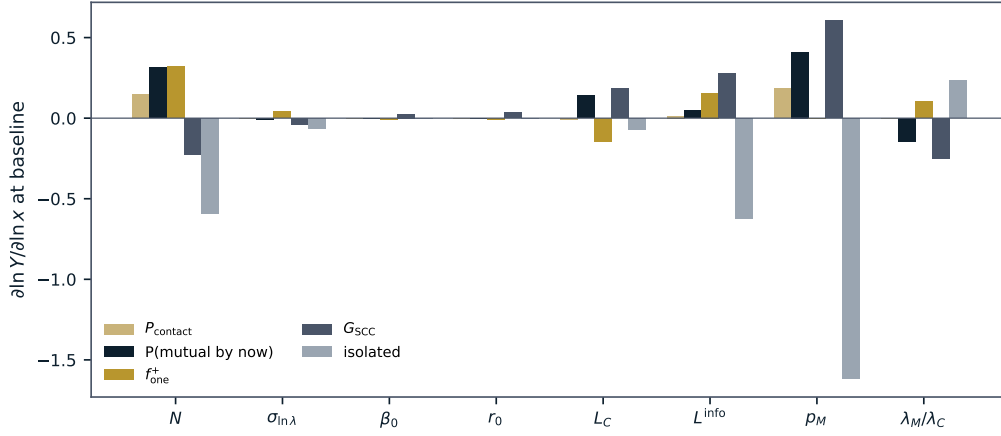


Figure 4: Local log-elasticities of the five graph statistics at the baseline (fixed-delay family), on $V(t)$, central differences of $\pm 20\%$ with common random numbers.

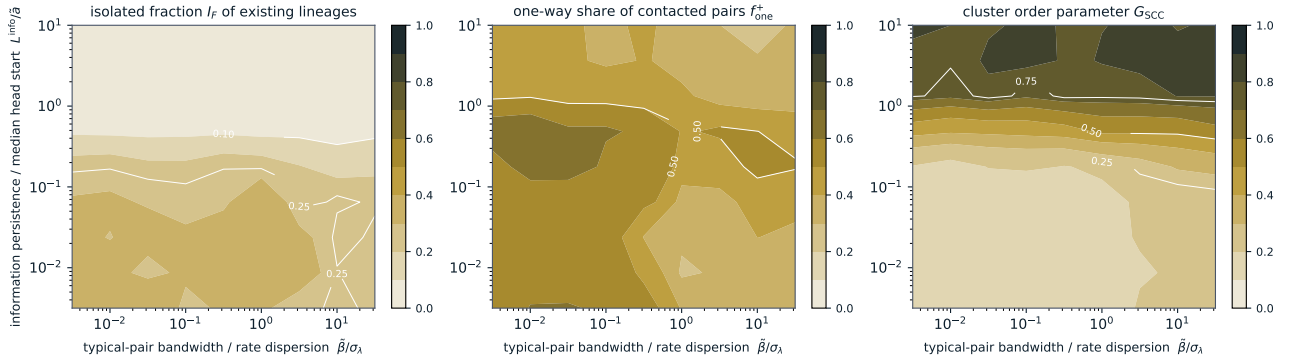


Figure 5: The persistence–bandwidth plane in dimensionless form, on $V(t)$: information persistence over the median head start of existing peers ($\tilde{a} = 2.4$ Gyr in the fixed-delay family) against typical-pair bandwidth over the standard deviation of rates ($\sigma_\lambda = 0.60$ for $\sigma_{\ln \lambda} = 0.5$; $\tilde{\beta} = 0.110 \beta_0$). Ten biological lineages of 50 Myr median lifetime, succession probability 0.5, synthetic lifetime held fixed at 1 Gyr so that only the information horizon varies, 100 runs per cell. Left: isolated fraction I_F of existing lineages. Middle: one-way share of contacted pairs f_{one}^+ . Right: cluster order parameter G_{SCC} .

the first numerical evidence for the conjecture in the form the phase diagram suggests, with intensity and persistence as the control parameters. It is not yet finite-size scaling: a collapse needs larger N , more runs, and a sweep in density, and the transition, if it is one, is not spatial, since the passive channel needs no bandwidth and the causal horizon of the box is a fraction of the head starts; what percolates is availability in time.

Active and passive channels. The strongest statement of a preliminary version of this analysis was that switching testimony off changes nothing, because persistent content is parsed by promotion alone, so that concealment is a property of the information horizon and not of the sender. That statement was an artefact of the channel definition, and it is withdrawn. In that version the bandwidth β_{ij} entered Postulate 1 only as a rate, accelerating the closing of a lead, while *access* to the other party’s content was assumed: an edge $i \rightarrow j$ formed as soon as j ’s content exceeded i ’s inside the causal cone, whether or not anything had passed between them. Height was conflated with access. The present model separates them. An edge needs the height condition *and* an open channel: the *active* channel, testimony while i is alive and growing, gated by a switch $a_{ij} \in \{0, 1\}$ and carrying the rate β_{ij} ; or the *passive* channel, observability of i ’s record after its death, open for the pair with probability $\kappa_{ij} = \min(1, \kappa_0 K(r_{ij}/r_0))$, after which j parses the record by promotion alone. The setting of that earlier version is $a \equiv 1$, $\kappa \equiv 1$, and every result above is for that setting. The four corners, at the baseline and at high persistence ($L^{\text{info}} = L_M = 5$ Gyr; 400 runs; `channels.py`), are:

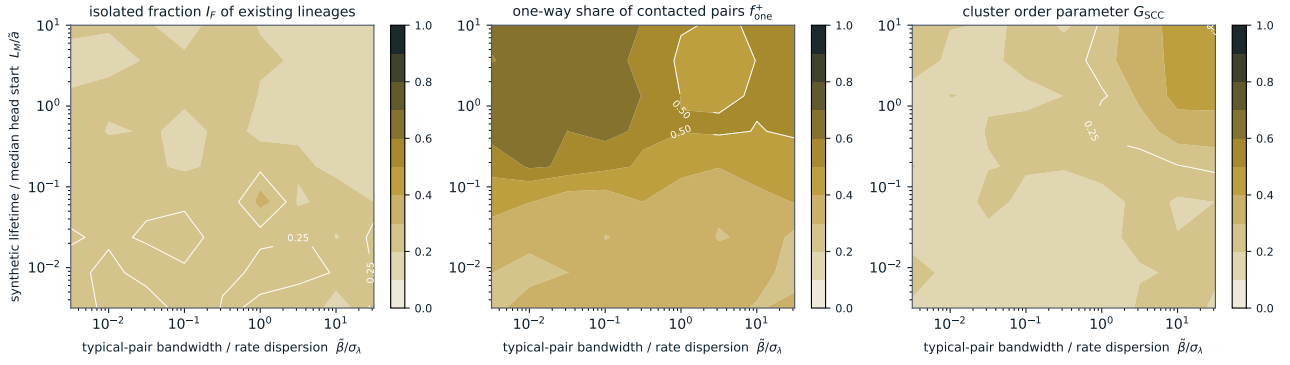


Figure 6: The same plane with the synthetic lifetime L_M varied instead, information persistence held at 0.5 Gyr. Persistence of an *active* lineage builds clusters only where bandwidth is also large; persistence of its *information* (Figure 5) does so at any bandwidth.

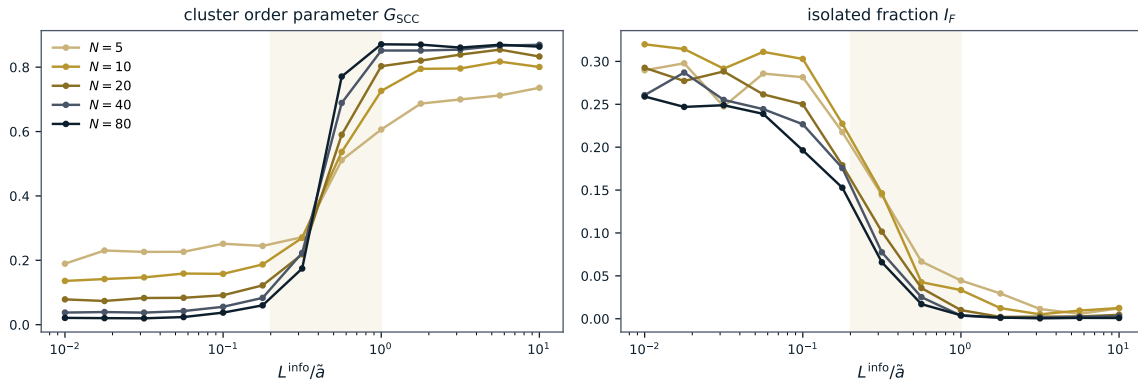


Figure 7: Finite-size behaviour of the persistence crossover: G_{SCC} (left) and I_F (right) against $L^{\text{info}}/\tilde{a}$ for $N = 5$ to 80 biological lineages at fixed density (box side $\propto N^{1/3}$), fixed $\beta_0 = 0.4$ and $r_0 = 150$ Mpc, $L_M = 1$ Gyr. The shaded band is the 0.2–1 range named in the text. The low plateau falls as $1/N$, the crossover sharpens, and its location drifts slowly towards $L^{\text{info}}/\tilde{a} \approx 0.4$.

	channels	P_{contact}	mutual	f_{one}^+	G_{SCC}	I_F
baseline	active and passive	0.97	0.86	0.64	0.20	0.21
	passive only ($a = 0$)	0.97	0.13	0.98	0.02	0.25
	active only ($\kappa = 0$)	0.93	0.67	0.73	0.13	0.32
	neither	0	0	–	0	1.00
	active, passive with $\kappa_0 = 1$	0.94	0.85	0.60	0.19	0.25
high persistence	active and passive	1.00	0.98	0.53	0.77	0.01
	passive only	1.00	0.96	0.68	0.50	0.01
	active only	0.94	0.70	0.73	0.13	0.30
	neither	0	0	–	0	0.99
	active, passive with $\kappa_0 = 1$	0.97	0.89	0.67	0.36	0.13

The channels do different work. Records alone (a silent civilisation leaving observable traces) sustain contact (P_{contact} unchanged) but of an almost wholly one-way kind, $f_{\text{one}}^+ = 0.98$ at the baseline, where mutual contact collapses to 0.13 because the synthetic lineages of the baseline are immortal and so never leave a record; at high persistence, where they do, records alone give mutual contact 0.96 but half the clustering. Testimony alone (active communication with no durable archaeology) gives mutual contact 0.67–0.70 and almost no clusters, $G_{SCC} = 0.13$, at either persistence, because clusters are built on the dead. Closing both is concealment, and it is total. A distance-dependent passive channel ($\kappa_0 = 1$, so a typical pair is observable with probability 0.11) costs little at the baseline and halves the clustering at high persistence. The corrected statement is

therefore this: in this model the character of contact is set by which channel is open, the one-way share by the passive channel and the clusters by persistence through it, and strategic darkness suppresses the active channel, and complete concealment additionally requires the passive channel to be absent or controllable. Whether a lineage can close its passive channel is a question about archaeology, not policy, and the model does not answer it.

5 FILTERS, DARKNESSES AND THE ACCESSIBLE INTENSITY

The paper's results assemble into one structure, and this section states it. The structure is more than a Fermi result: it is the framework statement that the preceding sections were built to test.

Five filters. Fix an observer o and a time t . A lineage with birth datum z (birth time, position, type) and mark m (rate, lifetime, succession, record persistence, channel gates) is *accessible* to o at t , $A_o(z, m; t) = 1$, when five conditions hold in order:

1. *existence*: the lineage has been born, $\tau \leq t$;
2. *persistence*: some content of it, as a living agent or as a record, still exists at a time when it could reach o ;
3. *causal reach*: that content lies in o 's past light cone;
4. *channel access*: an active channel (testimony while the sender lives) or a passive channel (observability of its record) is open to o ;
5. *cognitive access*: o 's content comes to contain it by t , which is the parse condition of Postulate 1.

The first three are conditions on spacetime and marks alone; the fourth is a condition on the pair; the fifth is a condition on o . A preliminary version of this analysis ran the first, second, third and fifth and let the fourth be implied by the fifth; the correction of Section 4 is what separates them, and once separated they do not recombine: a lineage may exist, persist, lie in o 's past and be within o 's cognitive reach if supplied, while no channel supplies it.

Proposition 5 (Generalised Fermi function). *Let origin lineages form a marked Poisson process on spacetime with intensity measure $\nu(dz)$ and independent marks, a successor being part of its origin's mark, and let the observer's own trajectory be fixed and unaffected by what it parses. Put $q_o(z, t) = \Pr[A_o(z, m; t) = 1 \mid z]$. Then the number of lineages accessible to o at t is Poisson with mean*

$$\Lambda_{\text{acc}}(o, t) = \int q_o(z, t) \nu(dz), \quad \text{and} \quad \Pr[\text{silence for } o \text{ at } t] = e^{-\Lambda_{\text{acc}}(o, t)}.$$

The Fermi function of Definition 1 is the special case $\nu = \rho dV \times (\text{uniform head starts}) \times f(\lambda) d\lambda$ with the first four filters taken as passed, so that q_o reduces to the cognitive factor $p_B(\lambda_0; \beta(r))$ and $\Lambda_{\text{acc}} = \rho V_{\text{eff}}$.

Proof. Condition on o as a point of the process; by the Slivnyak–Mecke theorem the remaining points are Poisson with the same intensity. Given z , $A_o(z, m; t)$ depends on the lineage's own mark and on o 's fixed trajectory, so the indicators of distinct lineages are independent, and independent thinning of a Poisson process by q_o is Poisson with intensity $q_o \nu$. Silence is the event that the thinned process is empty. ■

The scope of the proposition is exact and worth stating. It governs every statistic that is a sum over lineages of a per-lineage indicator evaluated by one observer: silence, the number of accessible peers, the first-contact time. It does not govern the graph statistics of Section 4, which are properties of the whole process and do not thin; that is why Section 2 is analytic and Section 4 is numerical. The classical Fermi argument reads a null observation as a statement about ν ; the proposition says it is a statement about Λ_{acc} , in which ν is multiplied by four factors that the argument does not see. V_{eff} is not a volume; it is the accessible measure of marked spacetime.

Darkness as filter failure. Each way of seeing nothing is the failure of one filter, and the correspondence is one-to-one once the fourth filter is separated from the fifth.

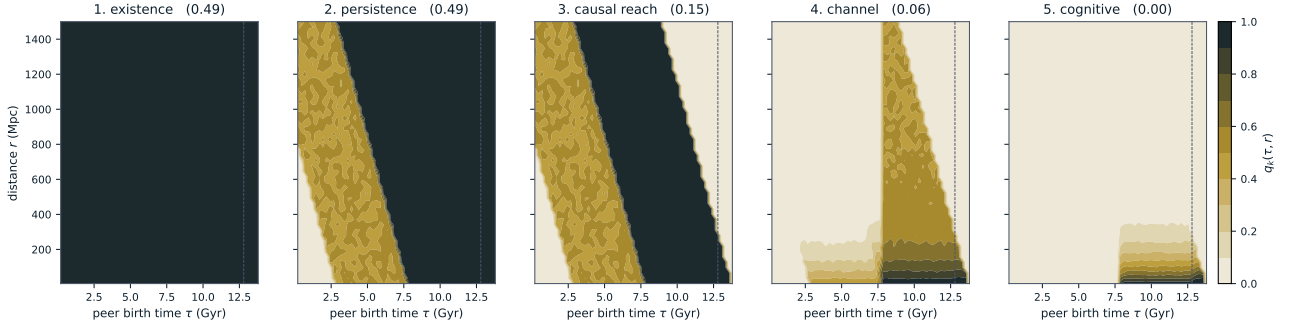


Figure 8: The five filters as nested regions of the (birth time, distance) plane for one observer at the present (median rate, born 1 Gyr ago, dashed line), in the fixed-delay family with the high-persistence setting ($L^{\text{info}} = L_M = 5$ Gyr) so that the regions are visible, and with a distance-dependent passive gate $\kappa_0 = 1$. Each panel is the probability $q_k(\tau, r)$ that a peer born at τ at distance r passes filters 1 to k , Monte Carlo over the peer's marks; the bracketed number is the fraction of a population uniform in volume to 1500 Mpc, with the family's birth density over the full horizon, that passes. The shell between successive panels is a darkness: emptiness between the full population and panel 1; temporal darkness between panels 1 and 3, where the slanted band is the light-cone coupling of age and distance; channel darkness between 3 and 4, where the near column is the passive channel and the recent column the active one; cognitive darkness between 4 and 5, where only recent and slow lineages are parseable by a young observer. Λ_{acc} is the intensity integrated over the last panel. At the baseline persistence the fractions are 0.49, 0.45, 0.11, 0.002 and below 10^{-3} .

filter that fails	name	mechanism in the model	where shown
existence	emptiness (not darkness)	ν small; the classical reading	q_F , Section 3
persistence or causal reach	<i>temporal</i> darkness	windows of existence, availability and reach fail to overlap	Figs 2, 5, 8
channel access, active	<i>strategic</i> darkness	$\beta \rightarrow 0$ by choice (Liu)	channel table, §4
channel access, passive	<i>archaeological</i> darkness	no record, or none observable: $L^{\text{info}} \rightarrow 0$ or $\kappa \rightarrow 0$	channel table; Figure 5
cognitive access	<i>cognitive</i> darkness	$\lambda_i - \lambda_j > \beta_{ij}$: the lead never closes	Prop. 1; f_{one}^+ : rate-multiple elasticity

Three remarks. The classical Fermi reading occupies one row, and it is the row that is not darkness. Liu's dark forest occupies one row, and the channel table shows that it conceals only together with the row below it: a civilisation may choose silence but cannot in general choose to leave no observable history, so strategic darkness suppresses the active channel and complete concealment additionally requires the passive one to be absent or controllable. And cognitive darkness is the only row that is one-sided by construction: the faster party sees the slower, so an observer in cognitive darkness is not thereby unobserved.

Two distinctions. The table rests on two distinctions that the Fermi literature conflates. The first is between a *population* and a *contactable population*: the chain existence $\rightarrow$ persistence $\rightarrow$ causal reach, measured by q_F , by $V(t)$, and by Λ_{acc} ; the silence floor of an earlier version of this analysis was the cost of conflating the first two. The second is between *information* and *access to information*: persistence $\rightarrow$ channel, measured by L^{info} against (a, κ) ; the concealment claim of a preliminary version was the cost of conflating these. Each distinction is split twice, which is why there are five filters and not three.

Controls. The numerical results reduce to three dimensionless ratios and the channel gates. The *exposure* $\Theta_F = NB/W$ of Proposition 4 decides whether enough finite-window opportunities are sampled for silence to be surprising; abundance and head start enter through it together, not as rival explanations. The *persistence ratio* $\Pi_P = L^{\text{info}}/\tilde{a}$ decides whether transient contacts stay isolated or accumulate into a persistent

connected structure, with the crossover near $\Pi_P \approx 0.4$ at the tested density and sharpening with population. The *rate-matching ratio* $\Pi_\beta = \tilde{\beta}/\sigma_\lambda$ decides the reciprocity of contact where channels are short, and little else. The gates (a, κ) decide whether information is reachable at all, and which character contact has. In one line: *exposure determines opportunity; persistence determines connectivity; bandwidth determines reciprocity, below the persistence crossover; channels determine observability*. Two cautions attach. Θ_F and Π_P are both window-to-age-offset ratios, but the window of the first is the receiver's budget and that of the second the sender's record, and the paper does not claim they are one parameter seen twice. And above the persistence crossover the one-way share is set by persistence, not bandwidth (the ridge of Figure 5), so the third clause carries its qualifier.

What is and is not new here. That the Fermi problem turns on observation rather than abundance is not new: Drake's L , the dissolution of [13], and the temporal-window arguments all cut between existing and observed. The contribution of this paper is the finer cut, channel access separated from cognitive access with the former split into active and passive, the composition of the five filters into one accessible intensity, and the phenomenology of the three ratios; and, negatively, the two conflation it found in its own earlier versions and the one it found in the bandwidth-centred form in which the conjecture was first stated in an earlier version of this series.

6 LIMITATIONS

The model is frozen at this version, and these are its limits as it stands. The spatial model is a static comoving box with r/c delays; the correct object is the FRW past light cone, $\chi = c \int dt/a(t)$, and V_{eff} with a stability measure $H_s(\mathbf{x}, t)$ inside the spatial integral. The opportunity and reset inputs are parametrisations in the spirit of the cited curves (Madau–Dickinson, Lineweaver, Behroozi–Peeples, Piran–Jimenez), not the curves themselves, the horizon is 25 Gyr so that late percentiles are not reportable, and the emergence hazard is fixed rather than integrated over a posterior given one technological biosphere. The rate model is a continuum postulate: contact at t_c means subsumption, the audit of a parsed peer's strategy costs further rungs [4, Section 5], the budget B is not yet related to N_{max} or lifetimes to the freeze-out of [4, Section 2], and the finite-time proximity that the discrete ladder would measure, the information distance on histories [5], is replaced by a lead in rungs. The channel gates are a switch and a Bernoulli draw where a physical model would have rates growing with testimony and with the record's size and age, and the statistics count a parse of a record as contact, so that contact with a lineage's current content has no statistic of its own. The population is varied as a count in a fixed box rather than drawn as $N \sim \text{Poisson}(\rho V)$ with volume separate. The finite-size sweep is evidence of sharpening, not a collapse. None of these is needed to establish what the paper claims, and each is a project rather than a repair.

7 OPEN PROBLEMS

1. **[Open] Accessibility percolation.** Prove Conjecture 1, or establish it by finite-size scaling: larger N , more runs, a sweep in density with $N \sim \text{Poisson}(\rho V)$, and a collapse of G_{SCC} against Π_P .
2. **[Open] Channels as rates.** Replace the gates (a, κ) by rates, separate contact with current content from contact with a record, and give strategic and archaeological darkness statistics of their own.
3. **[Open] Λ_{acc} with astrophysical inputs.** Compute the accessible intensity on the FRW light cone with cited opportunity and reset curves and a posterior over emergence, so that a null observation becomes a bound on Λ_{acc} with stated uncertainty.
4. **[Open] Parameter space.** The model has five primitive inputs $(\rho, f(\lambda), \beta(r), B)$, the gates (a, κ) and three structural choices (emergence family, lineage transition, persistence). Lay the space out explicitly, with the regions in which each conclusion holds, so that others can explore and criticise it.

8 CONCLUSION: FERMI'S PROBLEM AS AN ACCESSIBILITY PROBLEM

The model does not estimate the number of technological lineages in the Universe. It establishes a different point: population and observable population are not the same object, and the difference is not one filter but five. Between existence and observation lie persistence, causal reach, channel access and cognitive access, and for a marked Poisson population they compose into an observer-dependent accessible intensity Λ_{acc} with $\Pr[\text{silence}] = e^{-\Lambda_{\text{acc}}}$. A null observation is therefore evidence about the measure of the observer's accessible region in a marked causal-cognitive spacetime, and density, emergence history, persistence, channels and cognitive rate all contribute to that measure and cannot in general be separated by silence alone.

The analytic and numerical parts identify different controls at different filters. For short observational budgets the exposure NB/W governs, and gigayear head starts make finite-window silence common without a sparse population or a slow observer. In the marked cosmic-history model, persistence is the principal control on connectivity under the tested scenario measures: information that survives for an order-one fraction of the typical head start accumulates into a giant strongly connected component, with finite-size evidence that the crossover sharpens. Bandwidth alters the reciprocity of contact where channels are short and does not by itself create connectivity. Testimony and records are distinct channels, and eliminating one does not eliminate the other. These produce four routes to a dark observation, temporal, strategic, archaeological and cognitive, which need not coincide; silence is compatible with a populated environment, with one-sided observation, and with no concealment by anyone.

This is the quantitative endpoint of the programme of [2, 3, 4], whose synthesis is given there [2, Section 13]: across the four papers the one distinction is between information that exists and information that is effectively available to a particular observer, and limits, moduli, distance, persistence and channels are its successive forms. A synopsis of the programme is [1]; the thread continues in the open problems of Section 7, of which the first, the accessibility-percolation conjecture, is the one this paper leaves closest to decidable.

A dark universe, on this account, need not be empty, mutually invisible or hiding. It may be a universe in which the measure of what is physically present is much larger than the measure of what any finite observer can effectively access.

AUTHOR CONTRIBUTIONS

The question (whether the framework of [2] can say anything quantitative about the Fermi problem), the rate dynamics it builds on, and the proposal to generate head starts from cosmic history, substrate and stability windows are due to the author. The Fermi function and its closed form follow a proposal made in review of an earlier version, recomputed and corrected in the budget-to-head-start dependence; the marked-process formulation follows a critical reading, also made in review, of the author's proposal. The simulations, figures and text were produced by Claude, an AI system by Anthropic, under the author's direction, and every input is a scenario parametrisation that has not been replaced by cited curves. The simulation code is provided as ancillary files with the arXiv submission. The author takes full responsibility for all claims and errors; the model and its code were developed with an AI system and reviewed by the author, and have not been independently refereed.

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